

Fault-Onset Labeling Bounds Reported Accuracy in the Tennessee Eastman Benchmark: A Leakage-Controlled Study of Six Classical Classifiers

Abstract

Fault diagnosis in the Tennessee Eastman Process (TEP) is a standard benchmark for data-driven process monitoring, yet published results are difficult to compare. We present a leakage-controlled benchmark of six classical classifiers: logistic regression, decision tree, random forest, histogram-based gradient boosting, linear SVM and XGBoost. Data were partitioned by complete simulation runs, standardization was fitted on training data only, hyperparameters were selected on a separate validation partition, variability was measured over five seeds, and the frozen model was evaluated once on the complete official test partition of 10.08 million observations. XGBoost ranked first in repeated validation (macro-F1 0.7683 ± 0.0017) and reached a test macro-F1 of 0.7200, accuracy of 0.6754 and MCC of 0.6621 under the labels distributed with the dataset.

These figures are bounded by the benchmark itself. Each fault is injected 160 samples into every 960-sample test run, so literal labeling caps the recall of every fault class at 0.8333; faults 6 and 7 attain 0.8333 and 0.8332, the ceiling itself. Rescoring the same frozen model over the post-onset segment, without refitting, raises macro-F1 to 0.8054, accuracy to 0.7949 and MCC to 0.7852, and the apparent validation-to-test generalization gap of +0.0483 becomes -0.0125 under a symmetric comparison. Across seven classifiers the relative accuracy gain lies between 0.142 and 0.150 against a structural bound of 0.1587, so the effect is a property of the labeling convention rather than of any estimator. Normal operation and faults 3, 9 and 15 remain mutually confounded after the correction, consistent with their known weak observability.

Keywords: Tennessee Eastman Process, fault diagnosis, machine learning, process monitoring, reproducible benchmark, fault-onset labeling

31 1. Introduction

32 Industrial processes operate through interacting physical, chemical, and
33 control mechanisms. A local disturbance can propagate through feedback
34 loops, alter several measured variables, and eventually compromise product
35 quality, equipment integrity, safety, or environmental performance. Fault
36 detection and diagnosis (FDD) therefore constitute central elements of mod-
37 ern process monitoring [1, 2, 3]. Detection determines whether the process
38 has departed from acceptable operation, while diagnosis seeks to identify the
39 underlying condition so that operators or automated systems can respond
40 appropriately.

41 Model-based diagnosis can be effective when reliable first-principles mod-
42 els are available. However, plant-wide systems are frequently nonlinear, high-
43 dimensional, stochastic, and only partially observed. These properties have
44 motivated data-driven monitoring, in which statistical or machine-learning
45 models infer diagnostic boundaries from historical measurements [2, 4]. Clas-
46 sical multivariate methods such as principal-component analysis (PCA), par-
47 tial least squares (PLS), and Fisher discriminant analysis established impor-
48 tant foundations for process monitoring [5]. More recent studies increasingly
49 employ tree ensembles, support vector machine (SVM), neural networks, and
50 sequence models to represent nonlinear relationships and dynamic behavior.

51 The Tennessee Eastman Process (TEP), originally introduced by Downs
52 and Vogel [6], remains one of the most widely used benchmarks in this field.
53 It represents a reactor–separator–recycle chemical process with 41 measured
54 variables, 11 manipulated variables, and a diverse set of disturbances. Its
55 continued value arises from the coexistence of relatively distinctive faults
56 and difficult conditions whose trajectories overlap normal operation. The
57 expanded simulation data published by Rieth et al. [7] provide hundreds
58 of independent runs per condition and non-overlapping random seeds for
59 development and testing, enabling large-scale and reproducible evaluation.

60 Despite extensive use of the TEP, reported results are often difficult to
61 compare. Studies may split temporally adjacent observations rather than
62 complete runs, tune models on the test set, report only accuracy, or omit
63 class-level behavior. Treating autocorrelated observations as independent
64 experimental replicates can produce optimistic estimates, while a high global
65 score can hide failures in specific operating conditions. These issues are espe-
66 cially relevant when the normal state is confused with a fault, because false
67 alarms can reduce operator confidence and generate unnecessary interven-

tions.

This study addresses these limitations through a leakage-controlled benchmark of six classical supervised classifiers: logistic regression, decision tree, random forest, histogram-based gradient boosting, linear SVM, and XGBoost. The study asks five questions:

1. Which classical model provides the strongest macro-averaged diagnostic performance under repeated validation?
2. Does the selected model maintain its performance on a complete, independent official test partition?
3. Which operating conditions account for the remaining diagnostic errors?
4. How much of the reported error is attributable to the fault-onset labeling convention rather than to the classifier?
5. Which model offers the best compromise between diagnostic performance and computational cost?

The main contribution is an auditable protocol that separates model selection from final testing, preserves complete simulation runs, quantifies variability over five seeds, reports confidence intervals, and examines precision, recall, F1, and confusion patterns for all 21 classes. Rather than presenting a single headline score, the analysis identifies a compact ambiguity group that defines the practical limit of the selected classifier.

A second contribution is methodological and, in our view, of wider consequence for the TEP literature. The expanded dataset injects each fault after a fixed warm-up interval, so a fraction of every faulty run is nominally normal yet carries the fault label. This fraction differs between development runs (4%) and official test runs (16.7%). We quantify how much of the reported error and of the apparent generalization gap is produced by this convention alone, and we report every headline metric under both the literal labeling and a post-onset labeling. To our knowledge this decomposition is not routinely reported, which may account for part of the dispersion among published TEP results.

2. Related Work

2.1. Data-driven process monitoring

Data-driven FDD spans latent-variable modeling, classification, clustering, change detection, and temporal modeling. The review by Venkatasubramanian et al. [1] established a broad taxonomy covering quantitative models,

104 qualitative models, and process-history-based methods. Chiang et al. [4, 5]
 105 showed how PCA, discriminant PLS, and Fisher discriminant analysis could
 106 support chemical-process diagnosis. Later reviews emphasized that industrial
 107 data may be nonlinear, dynamic, nonstationary, multimodal, and autocorre-
 108 lated [2, 3]. These properties motivate flexible nonlinear classifiers, but also
 109 demand careful validation.

110 Multivariate statistical monitoring of the TEP has a long history. Raich
 111 and Çinar combined statistical process monitoring with disturbance diagnosis
 112 in multivariable continuous processes [25], and multiscale extensions using
 113 wavelet decompositions were introduced to separate phenomena occurring
 114 at different time scales [27]. A broader survey of plant-wide disturbance
 115 detection is given by Thornhill and Horch [30].

116 A substantial line of work addresses the serial correlation of process data
 117 directly. Dynamic PCA augments the data matrix with time-lagged variables
 118 to capture time-varying structure [40]; canonical variate analysis models the
 119 dynamics explicitly and was compared against PCA and DPCA on the TEP
 120 simulator by Russell et al. [41], later extended with kernel density estimation
 121 for nonlinear dynamics [42]. These methods are relevant here for a reason
 122 beyond their performance: they treat the temporal structure of a run as
 123 the object of analysis, whereas an observation-level classifier treats each row
 124 independently. The labeling question examined in Section 3.2 arises precisely
 125 at that boundary.

126 2.2. Classical machine learning for TEP diagnosis

127 The families evaluated here are standard: classification and regression
 128 trees [8], random forests [9], gradient boosting [10] and its regularized histogram-
 129 based formulation in XGBoost [11], together with multinomial logistic regres-
 130 sion and margin-based linear classification [12] as controls that test whether
 131 approximately linear decision surfaces suffice. All implementations in this
 132 study were based on scikit-learn [13], except XGBoost, which used its refer-
 133 ence implementation [11].

134 SVM have been applied to the TEP in several forms. Chiang et al. com-
 135 bined Fisher discriminant analysis with SVM for fault diagnosis [28]; Kulkarni
 136 et al. incorporated process knowledge into the kernel to detect TEP faults
 137 [29]; and kernel extensions of PCA and independent component analysis were
 138 compared against SVM on the same benchmark [31]. These studies use ker-
 139 nels that the present work could not afford at the sample size considered

140 here, which is the reason the linear surrogate of Section 3.4 should be read
141 as a lower bound rather than as the best attainable margin classifier.

142 Deep architectures have been applied extensively to the same benchmark.
143 Chadha and Schwung compared feedforward and convolutional networks for
144 TEP fault detection [36]; hybrid designs combining SVM with latent-variable
145 preprocessing [39], hierarchical long short-term memory (LSTM) structures
146 [37], and metaheuristically optimized deep models [38] have all reported
147 strong results. Most directly comparable to the present study, Lomov et
148 al. [35] evaluated recurrent, attention-based and temporal convolutional ar-
149 chitectures on the same expanded TEP dataset used here. The present work
150 does not propose a competing architecture; it examines a property of the
151 evaluation protocol that applies to all of them.

152 *2.3. Evaluation protocols and leakage*

153 Concern about evaluation protocol is not specific to process monitoring.
154 In a survey across disciplines that have adopted machine learning, Kapoor
155 and Narayanan identified leakage in seventeen fields, affecting at least 294
156 publications and in several cases producing conclusions that did not survive
157 correction [34]. Their taxonomy distinguishes eight mechanisms, two of which
158 bear directly on the present benchmark: temporal leakage, where information
159 from the future informs a prediction about the past, and non-independence
160 between training and evaluation samples, where records that share a common
161 origin are distributed across partitions and treated as independent replicates.

162 Both are latent in observation-level TEP classification. Consecutive sam-
163 ples within a simulation run are strongly autocorrelated, so a split performed
164 at the observation level places near-duplicate records on both sides of the par-
165 tition. The run-level protocol of Section 3.3 is designed to avoid this. The
166 labeling question analyzed in this paper is related but distinct: it concerns
167 not how records are divided, but whether the label attached to a record
168 describes the state of the plant at that instant.

169 *2.4. Evaluation beyond accuracy*

170 Accuracy can be misleading when performance varies substantially among
171 classes. Macro-F1 gives equal importance to every class by averaging class-
172 specific F1 scores, while weighted-F1 weights them by support. Balanced
173 accuracy averages class recalls. Matthews correlation coefficient (MCC) sum-
174 marizes agreement between predictions and true labels and remains informa-
175 tive for multiclass problems [14]. Confusion matrices are indispensable be-

Table 1: Treatment of the pre-onset window in observation-level studies on the Tennessee Eastman benchmark. Three studies on the same simulator adopt three distinct conventions: relabeling the pre-onset segment as normal, retaining it under the fault label, and not addressing it.

Study	Task	Methods	Pre-onset handling	Scored as fault
Yin et al. [32]	binary detection	SVM, PLS	stated, relabeled normal	800
Sun et al. [33]	detection + identif.	Bayesian RNN	stated, retained	960
Agarwal et al. [37]	21-class diagnosis	LSTM-SAE	not mentioned	960
This work	21-class diagnosis	six classical	stated, both protocols	960 / 800

176 cause they reveal whether errors are diffuse or concentrated in specific class
 177 pairs. The present study therefore uses macro-F1 as the primary model-
 178 selection criterion and treats the other metrics as complementary evidence.

179 2.5. Reported treatment of the fault-onset window

180 The analysis reported in this paper is only consequential if the treat-
 181 ment of the onset window varies, or goes unstated, across published work
 182 on the same benchmark. We therefore examined the data-description and
 183 experimental-protocol sections of studies for which the full text was avail-
 184 able, recording the diagnostic task, the stated treatment of the pre-onset
 185 segment, and the number of test samples scored against the fault label. The
 186 purpose is illustrative rather than exhaustive. Three studies on the same
 187 simulator adopt three different conventions, which is sufficient to establish
 188 that the choice is not standardized; Table 1 reports them.

189 The question applies to both distributions of the benchmark. In the
 190 original data of Downs and Vogel as distributed by Chiang et al. [4, 6],
 191 each condition is simulated for 24 hours in training and 48 hours in testing
 192 at a three-minute interval, giving 480 and 960 samples, with the fault in-
 193 jected after one hour and eight hours respectively. The pre-onset fractions
 194 are therefore 4.2% and 16.7%, essentially identical to those of the expanded
 195 dataset analyzed here. The artifact is thus a property of the benchmark
 196 design rather than of any particular redistribution of it, and the results of
 197 Section 4.5 bound what can be reported on either version.

198 Yin et al. [32] address the window directly. Their data description states
 199 that fault information emerges eight hours after start-up, so that the first 160
 200 test samples appear normal while the remaining 800 are genuinely faulty, and
 201 their discussion of the classification output is explicit about the consequence:
 202 the target label is -1 over the first 160 samples and $+1$ from the 161st

203 onward. The pre-onset segment is thus relabeled as normal rather than
204 scored against the fault, and their fault-detection rate is computed over 800
205 samples.

206 Sun et al. [33] state the same structure – the simulator runs 160 samples
207 in the normal state before the disturbance is introduced, then continues for
208 800 further samples – but evaluate over all 960. Awareness of the injection
209 instant is therefore not sufficient by itself; what determines comparability is
210 what is done with the segment.

211 Agarwal et al. [37] describe their data as 1440 samples per class, of which
212 the first 480 are used for calibration and the remaining 960 for testing, and
213 identify the source distribution explicitly. The instant of fault injection is
214 not mentioned anywhere in the data description or the experimental proto-
215 col. The consequence is not incidental to that study’s argument: its stated
216 objective is to improve the diagnosis of the incipient faults 3, 9 and 15, to
217 which end pseudo-random binary signals are actively injected into the plant.
218 Under literal labeling, part of the difficulty that motivates such an interven-
219 tion is contributed by observations in which no fault is yet present.

220 The pattern across the three is not one of carelessness, and it admits
221 a structural explanation. In binary detection the pre-onset segment has a
222 natural destination: the normal class already exists in the problem, so rela-
223 beling those observations is the obvious treatment, and the false-alarm rate
224 is in fact defined over precisely that segment. The convention is therefore
225 built into the evaluation. In multiclass diagnosis the same move is no longer
226 neutral, because assigning pre-onset observations of every faulty run to class
227 0 alters the prior of the normal condition and, in the present data, would
228 add 1.6 million observations to a class that already competes with faults 3,
229 9 and 15. Faced with that, the simplest course is to label each run by its
230 fault number throughout, and the distinction disappears without ever being
231 posed as a choice. What is lost in the transition from detection to diagnosis
232 is not information about the data but a convention for handling it.

233 The practical consequence is that reported figures are not commensurable
234 across the three treatments. For a single frozen model the difference between
235 literal and post-onset labeling is 0.0854 in macro-F1 and 0.1231 in MCC (Sec-
236 tion 4.5), larger than the margins that typically separate competing methods
237 on this benchmark. We do not claim a systematic review here; the point is
238 that three conventions coexist, that the choice is seldom identified as such,
239 and that it should be reported alongside the unit of partitioning.

240 3. Materials and Methods

241 3.1. Tennessee Eastman Process data

242 The TEP simulates a plant-wide reactor–separator–recycle system and
 243 includes 41 process measurements (XMEAS) and 11 manipulated variables
 244 (XMV), totaling 52 predictors [6]. The original problem statement specifies
 245 the plant but not a control structure; the simulations distributed as bench-
 246 mark data use the plant-wide control scheme of Lyman and Georgakis [26],
 247 so the recorded trajectories reflect both the process dynamics and the closed
 248 loops acting on them. The classification task contained 21 labels: class 0 rep-
 249 resented fault-free operation and classes 1–20 represented the predefined fault
 250 conditions. The CSV distribution used in this work is a format conversion
 251 of the expanded TEP simulation data of Rieth et al. [7, 15].

252 The four source files contained 15,330,000 observations and 55 columns:
 253 the 52 process variables plus the identifiers `faultNumber`, `simulationRun`,
 254 and `sample`. Development runs contained 500 observations each. Official
 255 test runs contained 960 observations each. Every class had 500 official test
 256 runs, yielding

$$21 \times 500 \times 960 = 10,080,000 \quad (1)$$

257 test observations. Each class therefore contributed 480,000 observations to
 258 the final test.

259 3.2. Fault-onset windows and labeling protocols

260 The expanded dataset is generated by simulating each condition for 25
 261 hours (development) or 48 hours (official test) with a sampling interval of
 262 three minutes, and by injecting the fault after a fixed warm-up interval: one
 263 hour in faulty development runs and eight hours in faulty test runs [7, 16, 17].
 264 In sample indices, the fault is active from sample 21 in development runs and
 265 from sample 161 in official test runs. Fault-free runs contain no injection.

266 Consequently, the labels distributed with the dataset are only condition-
 267 ally correct. In a faulty development run, samples 1–20 (4.0% of the run)
 268 describe nominal operation while carrying a fault label; in a faulty official
 269 test run, samples 1–160 (16.7% of the run) do so. Two consequences follow
 270 directly and are, to our knowledge, seldom made explicit.

271 First, under an observation-level protocol with literal labels, the recall of
 272 every fault class is bounded above by

$$R_c^{\max} = \frac{960 - 160}{960} = 0.8333, \quad c = 1, \dots, 20, \quad (2)$$

273 whenever the classifier assigns pre-onset observations to some class other than
 274 c . No such ceiling applies to class 0, whose test runs contain no injection.
 275 Second, because the mislabeled fraction differs between development (4.0%)
 276 and test (16.7%), a metric computed on the two partitions is not measuring
 277 the same quantity, and part of any observed validation-to-test degradation
 278 is attributable to this asymmetry rather than to generalization.

279 We therefore report all headline metrics under two labeling protocols:

- 280 • **Literal labeling (L)**: every observation retains the label distributed
 281 with the dataset. This is the protocol implicitly adopted by most pub-
 282 lished observation-level TEP results and is retained here for compara-
 283 bility.
- 284 • **Post-onset labeling (P)**: pre-onset observations of faulty runs are
 285 excluded *at evaluation time*, so that each scored observation carries a
 286 label consistent with the physical state of the plant. The model is not
 287 refitted: protocol P rescores the predictions of the same frozen model
 288 that was fitted under protocol L, including its 4% of contaminated
 289 training observations. The comparison L versus P therefore isolates
 290 the labeling convention alone, and the reported post-onset figures are
 291 conservative, since a model additionally trained without the contami-
 292 nated observations could only do better.

293 A third variant, in which pre-onset observations are excluded from fitting
 294 as well, was not evaluated here; it would confound the labeling effect with a
 295 change in the training distribution and is left for future work.

296 Protocol L is used for model selection, so that the frozen-model decision
 297 is unaffected by the present analysis. Protocol P is reported alongside it in
 298 Section 4.5. A third option, relabeling pre-onset observations as class 0, was
 299 not adopted because it alters the class prior of the normal condition and
 300 would confound the analysis of the ambiguity group.

301 3.3. *Leakage-controlled partitioning and preprocessing*

302 The original development runs were divided at the run level into training
 303 and validation partitions using a fixed random seed and a stratified 80/20
 304 split within each source and class. This produced 8,400 training runs (400

per class), 2,100 validation runs (100 per class), and 10,500 official test runs (500 per class). No run was assigned to more than one partition.

All 52 predictors were standardized using

$$z_{ij} = \frac{x_{ij} - \mu_j^{(\text{train})}}{\sigma_j^{(\text{train})}}, \quad (3)$$

where $\mu_j^{(\text{train})}$ and $\sigma_j^{(\text{train})}$ were estimated exclusively from the 4.2 million training observations. The fitted transformation was then applied unchanged to validation and test data. This procedure prevented information from the validation or test partitions from influencing preprocessing.

3.4. Candidate classifiers

Six model families were compared. Hyperparameter tuning evaluated three candidates per family, producing 18 configurations. The selected configurations were subsequently frozen:

- **Logistic regression:** $C = 1.0$, maximum 1,000 iterations;
- **Decision tree:** unrestricted depth and minimum leaf size 1;
- **Random forest:** 400 trees, unrestricted depth, 50% of features considered per split, and minimum leaf size 1;
- **HistGradientBoosting:** learning rate 0.05, 350 boosting iterations, and at most 31 leaf nodes;
- **Linear SVM:** stochastic-gradient hinge-loss classifier with $\alpha = 10^{-5}$, maximum 2,000 iterations, and tolerance 10^{-3} ;
- **XGBoost:** 250 trees, maximum depth 6, learning rate 0.1, row subsampling 0.8, column subsampling 0.8, multiclass probabilistic objective, and histogram tree construction.

The tuning stage ranked configurations by macro-F1 and used MCC as a secondary criterion. The official test partition was not accepted as an input by either the tuning or repeated-validation programs.

The linear SVM was implemented as a stochastic-gradient surrogate rather than as an exact margin solver, because quadratic-programming solvers were not tractable at this sample size. Its score should therefore be read as a lower

bound on linear-margin performance, and the evidence for nonlinearity reported in Section 5.1 rests primarily on the multinomial logistic regression, which was optimized to convergence. Kernel SVMs were not evaluated for the same computational reason.

3.5. Repeated validation

After selecting the best configuration for each model family, variability was measured across seeds 2026–2030. For each seed, 40 complete runs per class were sampled from training and independently from validation. Thus, every repetition used 420,000 training and 420,000 validation observations while retaining the full 500-observation trajectories of all selected runs.

For each metric, the mean, sample standard deviation, and 95% confidence interval were calculated across the five repetitions. With $n = 5$, the interval was

$$\bar{x} \pm t_{0.975,4} \frac{s}{\sqrt{5}}. \quad (4)$$

Because only five seeds were used, these intervals quantify observed seed-to-seed variability but should not be interpreted as definitive evidence of pairwise statistical superiority. In particular, overlapping intervals for the two leading models were interpreted conservatively.

In addition to the interval estimates, the six model families were compared with the Friedman rank test over the five seeds treated as blocks, followed by the Nemenyi post-hoc procedure [19]. Pairwise comparisons used the Wilcoxon signed-rank test with Holm correction. The critical difference of the Nemenyi procedure was computed as $CD = q_\alpha \sqrt{k(k+1)/6N}$ with $k = 6$ models and $N = 5$ blocks. Results are reported in Section 4.1. With five blocks these procedures have low power by construction, and they are reported to document that the prespecified ranking rule was not substituted for an inferential check, not to establish pairwise superiority.

Because a single frozen evaluation cannot express sampling uncertainty, the official-test metrics were additionally accompanied by 95% intervals obtained by resampling complete runs with replacement (1,000 bootstrap replicates over the 10,500 test runs). The run, not the observation, is the resampling unit, consistent with the argument that temporally adjacent observations are not independent replicates.

365 3.6. Frozen final evaluation

366 XGBoost candidate 1 was selected because it achieved the highest repeated-
367 validation macro-F1. The final seed was fixed at 2026 before test evaluation.
368 The final development set combined 40 complete runs per class from train-
369 ing and 40 from validation, totaling 840,000 observations. The frozen model
370 was then evaluated once on all 10,080,000 official test observations, without
371 subsampling or post-test adjustment.

372 Two earlier executions aborted during structural validation, before any
373 model fitting or metric computation, because of an incorrect run-length as-
374 sertion; only that assertion was changed, and the full incident log is available
375 in the repository.

376 The choice of 840,000 development observations, rather than the full
377 5,250,000 available, was imposed by the computational budget. To bound the
378 effect of that choice, the frozen configuration was refitted on $\{10, 20, 40, 80, 160\}$
379 complete runs per class and scored on the full validation partition, sampling
380 at the run level throughout. Results are reported in Section 4.2.

381 3.7. Performance measures and reproducibility

382 The reported measures were accuracy, balanced accuracy, precision, re-
383 call, class-specific F1, macro-F1, weighted-F1, MCC, confusion matrix, train-
384 ing time, inference time, inference time per observation, and serialized model
385 size. Zero divisions in class-specific measures were assigned zero. Data and
386 output integrity were monitored with SHA-256 hashes and manifests. The
387 final execution used Python 3.12.13, scikit-learn 1.9.0, and XGBoost 3.4.0 on
388 Linux.

389 4. Results

390 4.1. Model selection under repeated validation

391 Table 2 summarizes the five-seed validation results. XGBoost ranked first
392 with macro-F1 of 0.7683 ± 0.0017 and MCC of 0.7426 ± 0.0018 . HistGradi-
393 entBoosting was a close second, with macro-F1 of 0.7643 ± 0.0034 . Random
394 forest achieved 0.7398, followed by decision tree at 0.6714. The two linear
395 baselines performed substantially worse: logistic regression achieved 0.4685
396 and linear SVM achieved 0.4438.

397 The Friedman test over the five seeds rejected the hypothesis of equal
398 performance among the six families ($\chi^2_5 = 24.54$, $p = 1.71 \times 10^{-4}$), with a

Table 2: Repeated-validation performance over five seeds. Macro-F1 was the prespecified primary selection criterion.

Rank	Model	Macro-F1 (mean $\pm$ SD)	95% CI	MCC (mean $\pm$ SD)
1	XGBoost	0.7683 ± 0.0017	[0.7661, 0.7705]	0.7426 ± 0.0018
2	HistGradientBoosting	0.7643 ± 0.0034	[0.7601, 0.7686]	0.7391 ± 0.0045
3	Random Forest	0.7398 ± 0.0021	[0.7372, 0.7424]	0.7159 ± 0.0021
4	Decision Tree	0.6714 ± 0.0013	[0.6699, 0.6730]	0.6537 ± 0.0015
5	Logistic Regression	0.4685 ± 0.0074	[0.4592, 0.4777]	0.4567 ± 0.0073
6	Linear SVM	0.4438 ± 0.0099	[0.4314, 0.4561]	0.4377 ± 0.0109

399 Kendall coefficient of concordance of $W = 0.982$. The ordering is thus almost
400 perfectly consistent from seed to seed: XGBoost obtained the best macro-F1
401 in four of the five repetitions, and the mean ranks were 1.2 (XGBoost), 1.8
402 (HistGradientBoosting), 3.0 (random forest), 4.0 (decision tree), 5.0 (logistic
403 regression) and 6.0 (linear SVM).

404 Post-hoc resolution is nevertheless limited. With $N = 5$ blocks the Ne-
405 menyi critical difference is 3.372 mean ranks, so only three of the fifteen
406 pairwise comparisons are individually significant, all of them between a tree
407 ensemble and a linear baseline. The separation between XGBoost and Hist-
408 GradientBoosting is 0.6 mean ranks, far below the critical difference. The
409 Wilcoxon signed-rank test cannot help here either: with five paired observa-
410 tions its smallest attainable two-sided p -value is 0.0625, so no comparison can
411 reach $\alpha = 0.05$ even under a perfect win record, and after Holm correction
412 none does.

413 These two facts are complementary rather than contradictory. The high
414 concordance shows that the ranking is not an artifact of seed choice; the
415 wide critical difference shows that five repetitions cannot certify small dif-
416 ferences. Both support the conservative reading adopted here: the ordering
417 is reproducible, the gap between the two leading models is not resolvable at
418 this sample size, and additional seeds would be required to settle it.

419 The confidence intervals of XGBoost and HistGradientBoosting over-
420 lapped. Accordingly, XGBoost was selected by the prespecified macro-F1
421 rule, but the result is not presented as proof of a statistically significant
422 difference between the two models. The large separation between nonlin-
423 ear ensembles and linear baselines indicates that nonlinear interactions are
424 central to this diagnostic task.

Table 3: Validation macro-F1 as a function of the training set size, in complete runs per class. Δ is the gain over the preceding row, that is, the effect of doubling the data.

Runs/class	Observations	F1 (L)	Δ	F1 (P)	Δ	Fit (s)
10	105,000	0.7538	—	0.7780	—	59.0
20	210,000	0.7591	0.0052	0.7847	0.0067	65.2
40	420,000	0.7672	0.0082	0.7930	0.0083	111.6
80	840,000	0.7756	0.0083	0.7988	0.0058	175.4
160	1,680,000	0.7806	0.0051	0.8016	0.0028	357.4

4.2. Sensitivity to the size of the development set

Table 3 reports macro-F1 on the complete validation partition as a function of the number of complete runs per class used for fitting, under both labeling protocols. All fits used the frozen configuration and 24 threads.

Two observations follow. First, the procedure reproduces the repeated-validation result: at 40 runs per class it yields a macro-F1 of 0.7672 against the 0.7683 ± 0.0017 of Table 2, a difference of 0.0011, which confirms that the curve is measured under the protocol of Section 3.5 rather than under a variant of it.

Second, the returns diminish. Doubling the data from 80 to 160 runs per class adds 0.0051 in macro-F1 under literal labeling and 0.0028 under post-onset labeling; quadrupling from the 40 runs per class used in the frozen fit adds 0.0134 and 0.0086 respectively. The frozen model is therefore not at the asymptote, and the figures reported in Section 4.3 are conservative in that sense, but the residual headroom is of the same order as the separation between the two leading model families and does not alter the ranking.

The comparison that matters for this paper is between that headroom and the effect under study. Quadrupling the training data from the size actually used gains 0.0086 in post-onset macro-F1; changing the labeling convention on a single fixed model changes macro-F1 by 0.0854, an order of magnitude more. Whatever remains to be gained from more data, it is small beside what is gained or lost by how the pre-onset window is scored.

4.3. Official test performance

After the protocol was frozen, the selected XGBoost model was fitted to 840,000 development observations and evaluated on the complete official test set. Table 4 presents the final metrics under literal labeling. Accuracy and

451 balanced accuracy were both 0.6754, macro-F1 and weighted-F1 were both
 452 0.7200, and MCC was 0.6621. The equality of accuracy and balanced accu-
 453 racy, and of macro-F1 and weighted-F1, follows from the exactly balanced
 454 class supports, and holds only under literal labeling; under the post-onset
 455 protocol of Section 4.5 the fault classes retain 400,000 observations while the
 456 fault-free class retains 480,000, so the two pairs of metrics separate. Macro-
 457 F1 exceeds accuracy because accuracy equals macro-recall under balanced
 458 supports, while precision exceeds recall in most classes; the harmonic mean
 459 of a high precision and a moderate recall is larger than the recall alone. The
 460 gap between macro-F1 and accuracy is therefore itself a signature of the
 461 systematic under-recall documented in Section 4.5.

Table 4: Frozen XGBoost performance on the official test set.

Metric	Value
Accuracy	0.6754
Balanced accuracy	0.6754
Macro-F1	0.7200
Weighted-F1	0.7200
MCC	0.6621
Training time (s)	170.34
Inference time (s)	56.57
Inference (ms/observation)	0.005612
Development observations	840,000
Test observations	10,080,000

462 Relative to repeated validation, macro-F1 decreased by 0.0483 and MCC
 463 by 0.0805. Three mechanisms could account for this difference: independent
 464 random states in the test simulations, the longer duration of test runs, and
 465 the larger fraction of mislabeled pre-onset observations. The third mechanism
 466 is both the largest and the only one that is purely a labeling artifact. Under
 467 literal labeling the attainable recall per fault class falls from $480/500 = 0.96$
 468 in development runs to $800/960 = 0.8333$ in test runs, a ratio of 0.868, which
 469 is of the same order as the observed degradation. Section 4.5 shows that the
 470 gap does not merely shrink under post-onset labeling but changes sign: the
 471 post-onset test macro-F1 of 0.8054 exceeds the literal-label validation figure
 472 of 0.7683 by 0.0371.

473 That comparison alone would not be symmetric, because the validation
 474 figure was itself computed under literal labels and carries its own 4% pre-

onset contamination. The validation partition was therefore rescored under both protocols as well. Applying the selected XGBoost configuration to all 1,050,000 validation observations gave a literal-label macro-F1 of 0.7706, which agrees with the five-seed repeated-validation figure of 0.7683 to within 0.0023 and confirms that this partition is a faithful reference. Excluding the 40,000 pre-onset observations, 3.81% of the partition, raised macro-F1 to 0.7953, accuracy from 0.7556 to 0.7843, and MCC from 0.7440 to 0.7739.

The correction on the validation partition is thus +0.0246 in macro-F1, against +0.0854 on the test partition, a ratio of 0.29 that tracks the ratio of contaminated fractions (3.81% against 15.87%). Transferring the measured validation correction to the repeated-validation reference gives a post-onset validation macro-F1 of approximately 0.7929, against a post-onset test macro-F1 of 0.8054. Table 5 summarizes the decomposition.

Table 5: Validation-to-test difference in macro-F1 under both labeling protocols. The gap present under literal labeling does not survive a symmetric comparison.

Protocol	Validation	Test	Difference
Literal (L)	0.7683	0.7200	+0.0483
Post-onset (P)	0.7929	0.8054	−0.0125

Under a symmetric comparison the gap is not merely reduced but reversed, and its magnitude falls to roughly a quarter of the literal-label value, within the range of the seed-to-seed variability of Table 2. We therefore conclude that the reported generalization gap is an artifact of the labeling convention rather than evidence of degradation on unseen trajectories, and that the literal-label figure of 0.7200 understates the discriminative capacity of the frozen model by 0.0854 in macro-F1.

Two caveats attach to this conclusion. The validation rescaling used the tuned configuration fitted to the full training partition, not the five per-seed refits of the repeated-validation procedure, so the transferred correction is an estimate of the effect rather than an exact recomputation; the close agreement between 0.7706 and 0.7683 bounds the error of that substitution. Furthermore, the pre-onset destinations differ between partitions: 61.1% of pre-onset validation observations were assigned to the group $\{0, 3, 9, 15\}$ against 92.0% in the test partition, which is consistent with the development segment being only 20 samples long and overlapping the simulation transient.

Table 6: Class-wise results for the frozen XGBoost model on the official test set. Every class had support of 480,000 observations.

Class	Precision	Recall	F1	Class	Precision	Recall	F1
0	0.1481	0.2045	0.1718	11	0.8447	0.7095	0.7712
1	0.9894	0.8243	0.8993	12	0.9271	0.7669	0.8395
2	0.9896	0.8209	0.8974	13	0.9828	0.7444	0.8471
3	0.1636	0.4218	0.2357	14	0.9889	0.8187	0.8958
4	0.9477	0.8222	0.8805	15	0.1465	0.1978	0.1683
5	0.9879	0.8234	0.8982	16	0.9756	0.6681	0.7931
6	0.9973	0.8333	0.9080	17	0.9624	0.7809	0.8622
7	0.9964	0.8332	0.9075	18	0.9367	0.7760	0.8488
8	0.9665	0.7779	0.8620	19	0.7701	0.7559	0.7630
9	0.1328	0.2126	0.1635	20	0.8391	0.7029	0.7650
10	0.8066	0.6885	0.7429				

504 4.4. Class-specific performance

505 Table 6 demonstrates marked heterogeneity. Seventeen classes achieved
506 F1 scores from 0.7429 to 0.9080. Classes 6 and 7 were strongest, with F1 of
507 0.9080 and 0.9075, respectively. Classes 1, 2, 5, and 14 also approached or
508 exceeded 0.8958.

509 Four classes were persistently difficult: class 9 (F1=0.1635), class 15
510 (0.1683), normal operation (class 0; 0.1718), and class 3 (0.2357). Class
511 3 had the highest recall within this group (0.4218) but very low precision
512 (0.1636), indicating systematic overprediction.

513 A pattern in the recall column deserves emphasis. Classes 6 and 7 reached
514 recalls of 0.8333 and 0.8332, which coincide with the structural ceiling of
515 Eq. (2) to within one part in ten thousand. Classes 1, 2, 4, 5 and 14 lie
516 between 0.8187 and 0.8243, that is, at 98–99% of that ceiling. The frozen
517 model is therefore essentially saturated on these classes once the pre-onset
518 segment is removed, and their reported F1 values of roughly 0.88–0.91 are
519 limited by the labeling convention rather than by the classifier.

520 4.5. Effect of the fault-onset window

521 Table 7 compares the frozen model under the two labeling protocols of
522 Section 3.2. No refitting, retuning or reselection was performed; the same
523 frozen model and the same predictions are scored under two label conven-
524 tions.

Table 7: Frozen XGBoost under literal (L) and post-onset (P) labeling on the official test partition. Intervals are 95% run-level bootstrap intervals over 1,000 replicates.

Metric	Literal (L)	Post-onset (P)
Accuracy	0.6754 [0.6716, 0.6797]	0.7949 [0.7898, 0.8006]
Balanced accuracy	0.6754 [0.6748, 0.6760]	0.8006 [0.7998, 0.8013]
Macro-F1	0.7200 [0.7192, 0.7208]	0.8054 [0.8044, 0.8063]
MCC	0.6621 [0.6579, 0.6666]	0.7852 [0.7798, 0.7910]
Observations	10,080,000	8,480,000

525 Excluding the 1,600,000 pre-onset observations, 15.87% of the partition,
526 raised macro-F1 by 0.0854, accuracy by 0.1195 and MCC by 0.1231. No
527 model was refitted and no hyperparameter was changed.

528 Two diagnostics establish that this is a labeling effect rather than a
529 favourable choice of subset. First, the frozen model assigned 16.10% of the
530 pre-onset observations to class 0 and 75.86% to classes 3, 9 or 15, that is,
531 92.0% to the group of conditions that are indistinguishable from nominal op-
532 eration, and only 4.20% to the fault label they nominally carry. The model
533 therefore identifies the pre-onset segment as a period in which nothing de-
534 tectable is occurring, which is correct, and literal labeling scores this as error.
535 Second, restricting the evaluation to post-onset observations raised the recall
536 of the 17 well-separated classes to between 0.8015 and 0.9999, the upper value
537 corresponding to essentially complete recovery of faults 6 and 7, while the
538 ambiguity group improved only marginally (F1 of 0.2217, 0.3238, 0.2100 and
539 0.2106 for classes 0, 3, 9 and 15, against 0.1718, 0.2357, 0.1635 and 0.1683
540 under literal labels). The ambiguity group is therefore not a labeling artifact;
541 the ceiling on the remaining seventeen classes is.

542 The two phenomena are nevertheless connected. Class 3 was predicted
543 1,237,562 times against a true support of 480,000, an excess of 757,562 ob-
544 servations. A substantial part of that excess consists of pre-onset segments
545 belonging to other faults, which the model routes to the weakly observable
546 conditions because, at those instants, nothing has yet happened. The over-
547 prediction of class 3 is thus partly a consequence of the labeling convention
548 and partly a consequence of genuine overlap, and only the post-onset analysis
549 separates the two.

550 This distinction matters for comparability. As documented in Section 2.5,
551 published observation-level results on the expanded TEP dataset do not con-

Table 8: Effect of post-onset labeling across model families. The last column reports the relative accuracy gain $\Delta A/A_P$, whose structural upper bound is $1 - 8,480,000/10,080,000 = 0.1587$.

Model	Macro-F1 (L)	Macro-F1 (P)	$\Delta A/A_P$
Decision tree	0.6330	0.7167	0.148
Extra trees	0.6965	0.7786	0.148
Random forest	0.7081	0.7912	0.149
HistGradientBoosting	0.7128	0.7972	0.149
XGBoost [†]	0.7200	0.8054	0.150
Logistic regression	0.4283	0.4698	0.142
Linear SVM	0.4207	0.4655	0.142

[†]Frozen model of Table 4, complete test partition.

552 sistently state whether pre-onset samples were retained, discarded, or rela-
553 beled, yet the three choices are separated by several points of macro-F1 for an
554 identical model. Two studies reporting different scores may therefore differ
555 in labeling convention rather than in method. We recommend that the treat-
556 ment of the onset window be reported explicitly as part of the experimental
557 protocol, in the same way that the unit of partitioning is reported.

558 4.6. Generality of the labeling effect across model families

559 If the effect is a property of the labeling convention rather than of the
560 selected classifier, it should appear in every model evaluated on the same
561 partition. An auxiliary experiment was therefore rescored under both pro-
562 tocols. It predates the frozen benchmark, uses a per-class subsample of the
563 test partition and includes an extremely randomized trees model in place
564 of XGBoost, so only the differences between protocols are interpreted. No
565 model was refitted.

566 The five tree-based models gained between 0.0821 and 0.0854 in macro-
567 F1, a spread of 0.003 across estimators whose capacities differ by orders of
568 magnitude, from a single unpruned tree to a regularized boosting ensemble.
569 No overfitting mechanism produces that uniformity, because the degradation
570 associated with overfitting scales with model capacity.

571 The relative accuracy gain explains the pattern. If a classifier assigns
572 pre-onset observations to any class other than the labeled fault, its literal
573 accuracy is its post-onset accuracy scaled by the retained fraction of the par-
574 tition, so $\Delta A/A_P$ is bounded above by 0.1587 independently of the model.

575 The seven models fall between 0.142 and 0.150, the residual being the small
576 fraction of pre-onset observations that match the literal label by chance. The
577 linear baselines gain less in absolute terms, 0.0415 and 0.0448, only because
578 they start lower, and they route far fewer pre-onset observations to the group
579 $\{0, 3, 9, 15\}$, 20.6% and 41.9% against 65.7–70.2% for the ensembles, yet are
580 penalized in the same proportion. Any observation-level result reported on
581 this dataset under literal labels therefore understates the post-onset discrim-
582 inative capacity of the model by a factor predictable from its own accuracy.

583 4.7. Structure of the remaining errors

584 The normalized confusion matrix in Figure 1 shows that errors were not
585 uniformly distributed. They concentrated in the four-class group $\{0, 3, 9, 15\}$.
586 Among true class-9 observations, 32.39% were predicted as class 3, 20.24%
587 as class 0, and 18.08% as class 15. Normal observations were assigned to
588 classes 3, 9, and 15 in 32.02%, 21.51%, and 18.24% of cases, respectively. For
589 class 15, the corresponding proportions were 30.96%, 20.53%, and 20.03%
590 for predicted classes 3, 9, and 0.

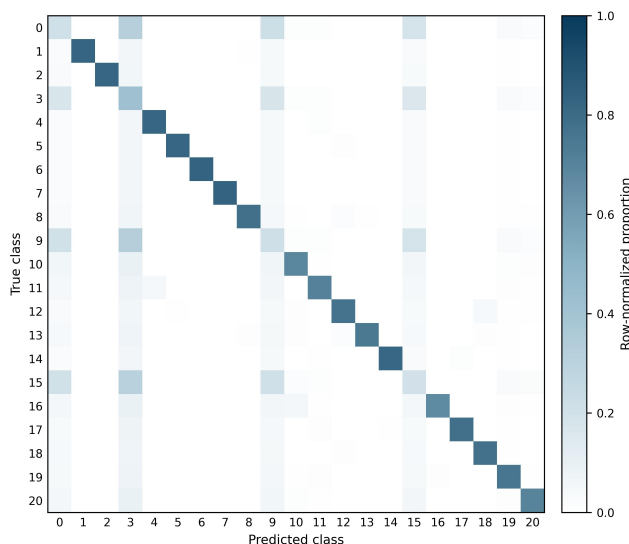


Figure 1: Row-normalized confusion matrix for the frozen XGBoost model on the official test partition.

591 Class 3 was predicted 1,237,562 times despite having 480,000 true obser-
592 vations. This excess explains its low precision and demonstrates that the

Table 9: Performance and computational cost of the six families under repeated validation (mean over five seeds, 420,000 training observations per repetition). Times measured on the environment of Section 3.7.

Model	Macro-F1	MCC	Train (s)	Inference (μ s/obs)	Size (MB)
XGBoost	0.7683	0.7426	864.7	7.03	18.9
HistGradientBoosting	0.7643	0.7391	73.5	29.40	16.9
Random Forest	0.7398	0.7159	477.3	20.29	8703.8
Decision Tree	0.6714	0.6537	54.0	0.24	28.8
Logistic Regression	0.4685	0.4567	65.4	0.15	0.005
Linear SVM	0.4438	0.4377	4.1	0.16	0.006

model used class 3 as a frequent destination for ambiguous trajectories. The persistence of the same difficult group during repeated validation and official testing suggests an intrinsic overlap under an observation-level representation rather than an isolated sampling artifact.

4.8. Computational performance

Table 9 reports the cost of all six families under identical conditions during repeated validation, which is required to answer the performance–cost question and cannot be inferred from the selected model alone. Training times are reported as means over the five seeds and are not directly comparable with the 170.34 s of Table 4, which was measured during the final evaluation on twice as many observations. The degree of parallelism was an invocation-time argument of each stage rather than a fixed setting, and it was not recorded in the run manifests, so the absolute times of the two stages reflect different thread counts on a 24-core host. The comparison between families within Table 9 is unaffected, because those six runs shared a single invocation, but the cross-stage comparison is not meaningful and no conclusion is drawn from it. Recording the degree of parallelism alongside the timing is a reproducibility requirement that this benchmark did not initially meet, and it has been added to the released code.

Final XGBoost training required 170.34 seconds. Prediction over 10.08 million observations required 56.57 seconds, corresponding to 0.005612 ms per observation or approximately 178,000 observations per second. The serialized final model occupied approximately 6.8 MB. These values indicate that the selected classifier is computationally feasible for high-throughput offline analysis and potentially for online scoring, although end-to-end deployment

latency must also include data acquisition, preprocessing, communication, and decision logic.

5. Discussion

5.1. Nonlinearity and model ranking

The ranking provides strong empirical evidence that the TEP classification boundary is not adequately represented by the evaluated linear models. XGBoost, HistGradientBoosting, and random forest substantially outperformed logistic regression and linear SVM even though all models received the same standardized predictors and run-level partitions. Tree ensembles can represent threshold effects and high-order interactions without requiring an explicit feature expansion, which is consistent with the nonlinear and feedback-driven nature of the TEP.

XGBoost and HistGradientBoosting differed by 0.0040 in mean macro-F1 with overlapping intervals, so the defensible conclusion is that XGBoost won under the prespecified rule, not that it is superior under all resampling regimes. Table 9 shows that the practical choice does not favour either: HistGradientBoosting trained 11.8 times faster, while XGBoost inferred 4.2 times faster, so the decision follows from whether the deployment is retraining-bound or scoring-bound. Model size is a third and largely independent constraint: the random forest required some 460 times the XGBoost artifact for 0.0285 less macro-F1, which rules it out for embedded deployment without being visible in any predictive metric. None of these conclusions can be drawn when only the selected model is profiled.

5.2. Why global performance is insufficient

The official macro-F1 of 0.7200 summarizes useful performance, but it does not describe a uniformly reliable 21-class diagnostic system. If only macro-F1 or accuracy had been reported, the strong results for 17 classes would obscure the near-failure of normal operation and faults 3, 9, and 15. In operational terms, low precision for normal operation and extensive exchange between normal and faulty labels can increase false alarms and missed interventions.

This result illustrates why class-level precision, recall, F1, and confusion structure should accompany aggregate measures in industrial diagnosis. It also supports a distinction between *benchmark discrimination* and *deployment readiness*. The present model is a strong benchmark classifier for most

653 faults, but additional safeguards are necessary before it could support au-
654 tonomous decisions.

655 *5.3. Implications for method selection in industrial practice*

656 Benchmark figures circulate as evidence of capability: a method reporting
657 a macro-F1 of 0.80 on the TEP appears preferable to one reporting 0.72.
658 Sections 4.5 and 4.6 show that a difference of exactly that size separates two
659 labeling conventions applied to an identical model. The question to put to a
660 reported figure is therefore not only which algorithm produced it, but over
661 which observations it was computed.

662 The fixed injection instant of the benchmark has no counterpart in an
663 operating plant, but the interval it stands for does. A disturbance entering
664 a unit does not become visible in the instrumentation at the moment it
665 occurs. It propagates through transport lag, and it is actively opposed by the
666 control loops, whose rejection of the disturbance is precisely what conceals it:
667 the difficulty of detecting faults inside chemical-process control loops due to
668 feedback masking is long recognized [20]. For incipient conditions the margin
669 is narrow to begin with, with reported amplitudes on the order of one to ten
670 per cent of the fault-free signal [21], so the deviation may remain within
671 normal variability for an extended period. The interval between occurrence
672 and annunciation is itself a design quantity in alarm management, where
673 deadbands and delay timers trade detection delay against nuisance alarms
674 [22]. Whatever label a historical dataset attaches to observations in that
675 interval, a model receives data carrying no signature of the condition it is
676 expected to report, and a figure of merit computed over them measures the
677 labeling convention rather than the diagnostic system.

678 The group $\{0, 3, 9, 15\}$ makes the same point in a sharper form. These
679 conditions survive the correction of Section 4.5 because they leave no persis-
680 tent signature in the 52 measured variables. For a deployment this is not a
681 modeling deficiency to be repaired with a better classifier; it is a statement
682 about what the installed instrumentation can observe. The literature treats
683 this as a design problem rather than an algorithmic one: sensor location can
684 be posed explicitly against fault-diagnostic observability criteria [23], and a
685 comprehensive formulation combining diagnosability, reliability and sensor
686 cost has been demonstrated on the Tennessee Eastman flowsheet itself [24].
687 The implication for practice is that when a condition proves undiagnosable
688 from the available measurements, the productive response is to revisit the
689 measurement set, not to continue tuning the classifier.

690 The cost results of Table 9 show that no model dominates. HistGradient-
691 Boosting trained 11.8 times faster than XGBoost for a difference of 0.0040
692 in macro-F1, while XGBoost inferred 4.2 times faster; the random forest re-
693 quired approximately 8.7 GB. Which constraint binds is a property of the
694 installation rather than of the benchmark: a model retrained weekly on a
695 workstation and a model scoring within the scan cycle of a controller are
696 limited by different quantities, and neither limit is visible in a predictive
697 metric.

698 The common thread is that benchmark performance and deployment
699 readiness are distinct properties, and that the distance between them is not
700 only a matter of model quality. It also depends on how the evaluation was
701 constructed, on what the instrumentation can observe, and on the resources
702 available where the model will run.

703 *5.4. Interpretation of the ambiguity group*

704 The identity of the difficult group is not a new finding, and it would be
705 misleading to present it as one. Faults 3, 9 and 15 have been reported as
706 weakly detectable since the earliest systematic studies of the TEP: moni-
707 toring statistics based on PCA, discriminant PLS and Fisher discriminant
708 analysis report missed-detection rates close to those of nominal operation
709 for exactly these conditions [4, 5], and later comparative studies reproduce
710 the pattern across a wide range of data-driven statistics [18]. The contribu-
711 tion here is the scale and the protocol: the same group is recovered under
712 run-level partitioning, on 10.08 million independent test observations, and it
713 persists after the labeling artifact of Section 4.5 is removed. This strengthens
714 the interpretation that the overlap is a property of the measured variables
715 rather than of any particular estimator.

716 The mechanism, however, constrains which remedies can work. If these
717 disturbances leave no persistent signature in the 52 measured variables, then
718 temporal aggregation over a run reduces the variance of an estimate that is
719 already centered on an ambiguous region: it will sharpen the decision without
720 moving it toward the correct class. Run-level aggregation should therefore
721 be expected to help the 17 separable classes, whose pre-onset transients are
722 transient, far more than the group $\{0, 3, 9, 15\}$, and this asymmetry is itself a
723 testable prediction. A hierarchical design that isolates the ambiguous group
724 and applies a discriminator with explicit memory to classes 0, 3, 9 and 15
725 is the more promising direction, but its plausibility rests on whether any
726 signature exists at all; establishing this is a question of observability, not

727 of classifier capacity, and it is better addressed with residual-based or first-
728 principles analysis than with additional supervised tuning.

729 Both proposals arise from test-set diagnosis and must therefore be eval-
730 uated under a newly defined development/test protocol; the present official
731 test must not be reused for tuning them.

732 5.5. *Threats to validity*

733 The TEP is a high-fidelity simulation, not a physical plant, so the results
734 need not transfer to sensor drift, maintenance changes, unmodeled distur-
735 bances or evolving operating modes; and the evaluation is pointwise mul-
736 ticlass diagnosis, which does not measure detection delay, false alarms per
737 run or sustained alarm logic. The uncertainty estimates rest on five seeds
738 and characterize the implemented experiment rather than all possible parti-
739 tions, and the hyperparameter search was deliberately limited to three can-
740 didates per family, so individual algorithms, particularly the linear baselines,
741 might improve under a wider search. The linear SVM in particular was a
742 stochastic-gradient surrogate and should not be read as the best attainable
743 margin classifier.

744 Three further caveats bear on quantities reported here. The final model
745 used 840,000 of the 5,250,000 available development observations and selec-
746 tion was performed at 420,000; Section 4.2 bounds the residual headroom at
747 0.0086 in post-onset macro-F1 under quadrupling, so the reported figures are
748 conservative, but a ranking established at one scale need not be invariant to
749 it. The degree of parallelism was not recorded in the run manifests, so the
750 times in Table 9 support ordinal comparison between families, which shared
751 one invocation, but not comparison against other stages. Finally, class sup-
752 port is balanced by construction, whereas industrial faults are rare; metrics,
753 thresholds and alarm costs would require recalibration under field prevalence.
754 Interpretation throughout focused on output behaviour and did not include
755 feature attribution or causal analysis.

756 6. Conclusion

757 This work presented a leakage-controlled benchmark of six classical clas-
758 sifiers for 21-class fault diagnosis in the Tennessee Eastman Process, using
759 run-level splitting, train-only standardization, separate tuning and valida-
760 tion, five-seed repetition and a single frozen test evaluation.

761 XGBoost ranked first in repeated validation (0.7683 ± 0.0017) and pro-
762 duced a test macro-F1 of 0.7200, accuracy of 0.6754 and MCC of 0.6621 over
763 10.08 million observations under literal labeling, rising to 0.8054, 0.7949 and
764 0.7852 once the pre-onset segment is excluded, with no refitting. Rescoring
765 the validation partition under the same protocol turns an apparent general-
766 ization gap of +0.0483 into -0.0125 . Several of the seventeen well-diagnosed
767 classes were shown to sit at the structural recall ceiling imposed by the la-
768 beling convention rather than at the limit of the classifier, while normal
769 operation and faults 3, 9 and 15 remain confounded after the artifact is re-
770 moved.

771 Three conclusions follow. Nonlinear tree ensembles diagnose most TEP
772 conditions effectively. Aggregate scores overstate practical reliability when
773 a small group of classes dominates the remaining errors. And a benchmark
774 result on these data is not comparable across studies unless the treatment
775 of the fault-onset window is stated: a single frozen model differs by 0.085 in
776 macro-F1 between two defensible conventions, more than most improvements
777 reported between competing architectures, and the effect is common to all
778 seven model families tested. We recommend that the treatment of the onset
779 window be reported as part of the experimental protocol, alongside the unit
780 of partitioning.

781 Future work should investigate temporal aggregation, hierarchical diag-
782 nosis, probability calibration and cost-sensitive alarm rules under a new
783 untouched test protocol. Sequence models raise the question in a richer
784 form: windowed labeling introduces a third category of observations, those
785 spanning the injection instant, whose proportion depends on the windowing
786 scheme of each study.

787 Data and Code Availability

788 The expanded Tennessee Eastman simulation data are publicly available
789 from Harvard Dataverse [7]. The CSV conversion used in this study is avail-
790 able through Kaggle [15]. All code, configuration files, run-level split mani-
791 fests, frozen hyperparameters, five-seed validation outputs, final predictions,
792 class-level metrics, confusion matrices, SHA-256 data hashes, and the envi-
793 ronment specification are deposited in a public archive with a reserved DOI.
794 The repository URL and the DOI identify the authors and are therefore with-
795 held from this anonymized version; both are cited in the accepted version,
796 and the deposit can be released to the editor on request during review. The

797 analysis of the fault-onset window can be reproduced from the deposited
798 predictions without refitting any model.

799 CRediT Authorship Contribution Statement

800 **First author:** Conceptualization, Investigation, Validation, Writing –
801 original draft, Writing – review & editing. **Second author:** Investigation,
802 Validation, Writing – original draft, Writing – review & editing. **Third**
803 **author:** Conceptualization, Methodology, Software, Formal analysis, Data
804 curation, Writing – original draft, Visualization, Supervision, Project admin-
805 istration.

806 Declaration of Competing Interest

807 The authors declare that they have no known competing financial inter-
808 ests or personal relationships that could have appeared to influence the work
809 reported in this paper.

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